EXPERIMENT 6: Measurement of AC Circuit Power and Power Factor Improvement

INTRODUCTION

The apparent power will be larger than P if the power factor is less than 1. Thus the current I that must be supplied will be larger for pf < 1 than it would be for pf=1, even though the average power P supplied is the same in either case. A larger current cannot be supplied without additional cost to the utility company. Thus, it is in the power company, best interest that major loads on the system have power factor as close as 1 as possible. In order to maintain the power factor as close to unity, power companies install banks of capacitor throughout the network as needed. They also impose as additional charge to industrial consumers who operate at low power factor. Since industrial loads are inductive and have low lagging power factors, it is beneficial to install capacitor to improve the power factor.

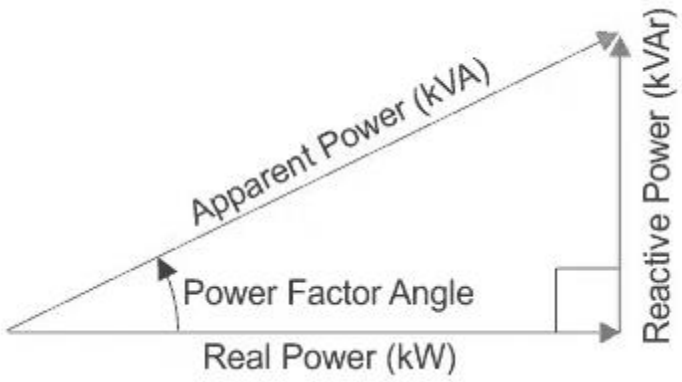


Figure 1: Power factor triangle

$$\cos\phi = \frac{Active\ power}{Apparent\ power}$$

Need for Power Factor Improvement

- Real power is given by P = VIcosφ. The electrical current is inversely proportional to cosφ for transferring a given amount of power at a certain voltage. Hence higher the pf lower will be the current flowing. A small current flow requires a less cross-sectional area of conductors, and thus it saves conductors and money.
- From the above relation, we see having a poor power factor increases the current flowing in a conductor, and thus copper loss increases. A large voltage drop occurs in the alternator, electrical transformer, and transmission, and distribution lines – which gives very poor voltage regulation.
- The KVA rating of machines is also reduced by having a higher power factor, as per the formula:

$$KVA = \frac{KW}{cos\phi}$$

Hence, the size and cost of the machine are also reduced.

The reason why we need to maintain high power factor;

- No additional charges in monthly electricity bill as there is no Power Factor Surcharge
- Extends the lifespan of electrical appliances
- Reduces electricity wastage for electrical installations at customers' premises
- Conserves the environment by lowering fuel usage and helps the country reduce carbon dioxide (CO2) emissions

OBJECTIVE

To study the method and significance of improving power factor.

EQUIPMENT AND COMPONENT

1. Digital AC Voltmeter, AC Ammeter, AC Watt hour meter and power factor meter.
2. HL2 (Fluorescent lamp base – 2 unit), HL3 (Fluorescent lamp starter base – 1 unit), 8W tube, starter, L4 (8W fluorescent lamp ballast), C09 (Capacitor, 2uF).

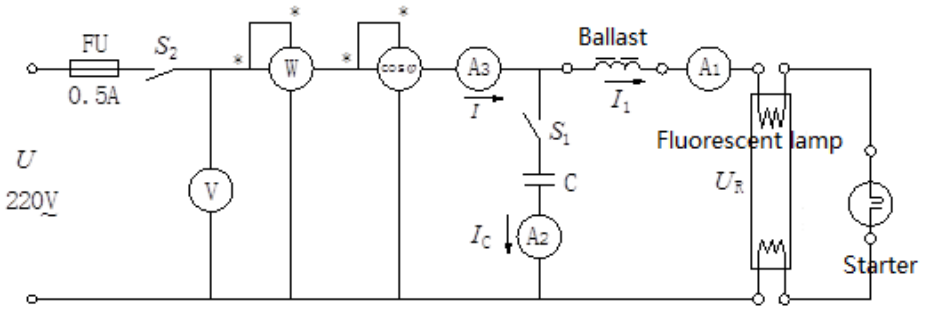


Figure 1: Power factor improvement circuit

PROCEDURE

1. Wire up the circuit as shown in Figure 1.
2. Set the voltage supply to AC 220V by using AC power supply.
3. **Double-check your connection with your lecturer before TURN ON power supply.**
4. After confirm your connection, turn on supply.
5. Start with 0uF capacitor and repeat for 1.5uF and 2uF.
6. Measure and records all parameter in Table 1.

RESULT

Table 1: improving the experimental data of line power factor

No.	Compensation capacitor C（μ F）	Total voltage U（V）	Total current I（mA）	Fluorescent lamp current I_1（A）	Reactive Power, Q (Var)	Real Power P（W）	Power factor $\cos\varphi$
1	0						
2	1.5						
3	2.0						

DISCUSSION

1. Elaborate your result and findings in this experiment.
2. Why is important to maintain high power factor?